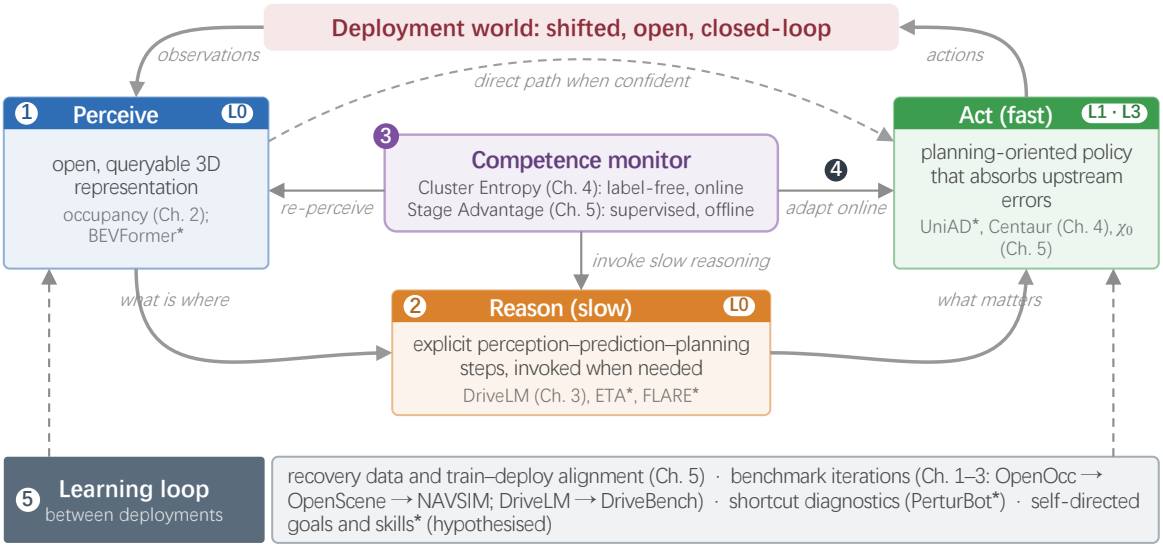

Deployment world: shifted, open, closed-loop
observations
actions
direct path when confident
1 Perceive L0
open, queryable 3D representation
occupancy (Ch. 2); BEVFormer*
3 Competence monitor
Cluster Entropy (Ch. 4): label-free, online
Stage Advantage (Ch. 5): supervised, offline
4
Act (fast) L1 · L3
planning-oriented policy that absorbs upstream errors
UniAD*, Centaur (Ch. 4), χ_0 (Ch. 5)
re-perceive
adapt online
invoke slow reasoning
what is where
what matters
2 Reason (slow) L0
explicit perception–prediction–planning steps, invoked when needed
DriveLM (Ch. 3), ETA*, FLARE*
5 Learning loop
between deployments
recovery data and train–deploy alignment (Ch. 5) · benchmark iterations (Ch. 1–3: OpenOcc → OpenScene → NAVSIM; DriveLM → DriveBench) · shortcut diagnostics (PerturBot*) · self-directed goals and skills* (hypothesised)